Introduction:

Gene expression can be studied to determine risk for certain diseases. Many proteins are considered biomarkers; their level of expression can indicate disease. CD99 is a transmembrane protein that functions in cell migration, death, and differentiation. It is specifically part of the leukocyte-endothelial transmigratory pathway. CD99 on both the leukocyte as well as the endothelial cells must bind in order for diapedesis to occur (1). Because of its role in transmigration its overexpression is common in many cancers including Ewing Sarcoma, lymphoblastic lymphoma/leukemia, and malignant gliomas, where it allows migration, invasion, and metastasis (2).

Our project goal was to analyze the effect of gender on tissues in the frontal cortical regions of the brain. Our dataset consisted of the microarray expression profiles from Affymetrix of 41 brain tissue samples extracted post-mortem (3).

Methods:

Workflow:

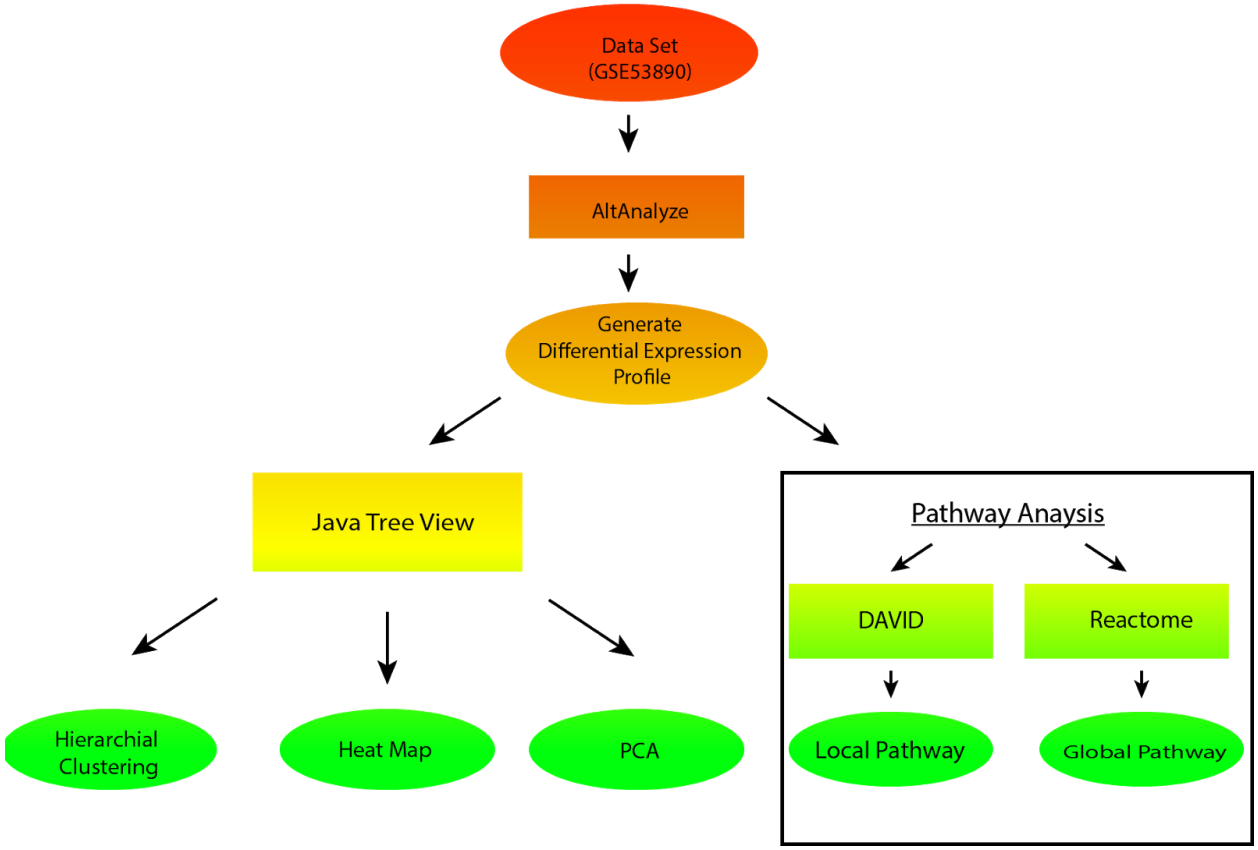

Figure 1. Methodology flow chart

Analysis of Differentially Expressed Genes:

Microarray expression profiles of the brain tissues were analyzed using AtlAnalyze v 2.1.0. Our data set was split into males and females: 21 males and 20 females. We used default settings for the analysis fold change 2. This gave us the expression profile matrix which was used to find a list of upregulated and downregulated genes.

Control of False Positives:

Benjamini-hochberg was used to control the false discovery rate.

Clustering:

AltAnalyze was used to create a hierarchial clustering heat map. Java TreeView was used to export image files of the clustering results. The column clustering method used was hopach and the row clustering method used was weighted. Rows were normalized relative to the row median.

Principal Component Analysis:

Principle Component Analysis was used to visualize trends. This was done in AltAnalyze on the total data set and the data was displayed on a three-dimensional map. The False Discovery Rate was set at 0.05 and the fold change was 2.

Pathway Analysis:

DAVID was used to find pathway analysis results. We used the Kegg pathway. Reactome was used to find the global picture of our pathway analysis results.

Results and Discussion:

The data set was segregated into two comparison groups based on sex. Age was not taken into account. Since the two groups both consisted of brain tissue from *Homo sapiens*, it was not necessary to include a 3rd control that would essentially function as a negative control.

Table 1 lists the differentially expressed genes between the two comparison groups. All of the upregulated genes, bar two are Y-linked, which is intuitive. Given that females do not have a Y chromosome, it is expected that all the Y-linked genes would be upregulated when compared with a female comparison group. Therefore, attention was paid to the two genes that were not Y-linked: Lysine Demethylase 5D and CD99. The down regulated genes are all X inactivated, and again, given that we are comparing two groups with different sex genotypes, this is also in accordance with what we expect.

Differential Genes:

Upregulated Genes	Downregulated genes
DEAD-box helicase 3, Y-linked	X inactive specific transcript (non-protein coding) - 214218_s_at
ubiquitin specific peptidase 9, Y-linked	X inactive specific transcript (non-protein coding) - 221728_x_at
ribosomal protein S4, Y-linked 1	X inactive specific transcript (non-protein coding) - 224589_at
eukaryotic translation initiation factor 1A, Y-linked	X inactive specific transcript (non-protein coding) - 224588_at

neuroligin 4, Y-linked	X inactive specific transcript (non-protein coding) - 227671_at
lysine demethylase 5D	X inactive specific transcript (non-protein coding) - 224590_at
ubiquitin specific peptidase 9, Y-linked	
CD99 molecule (Xg blood group)	

Table 1. List of Differentially Expressed Genes

Clustering:

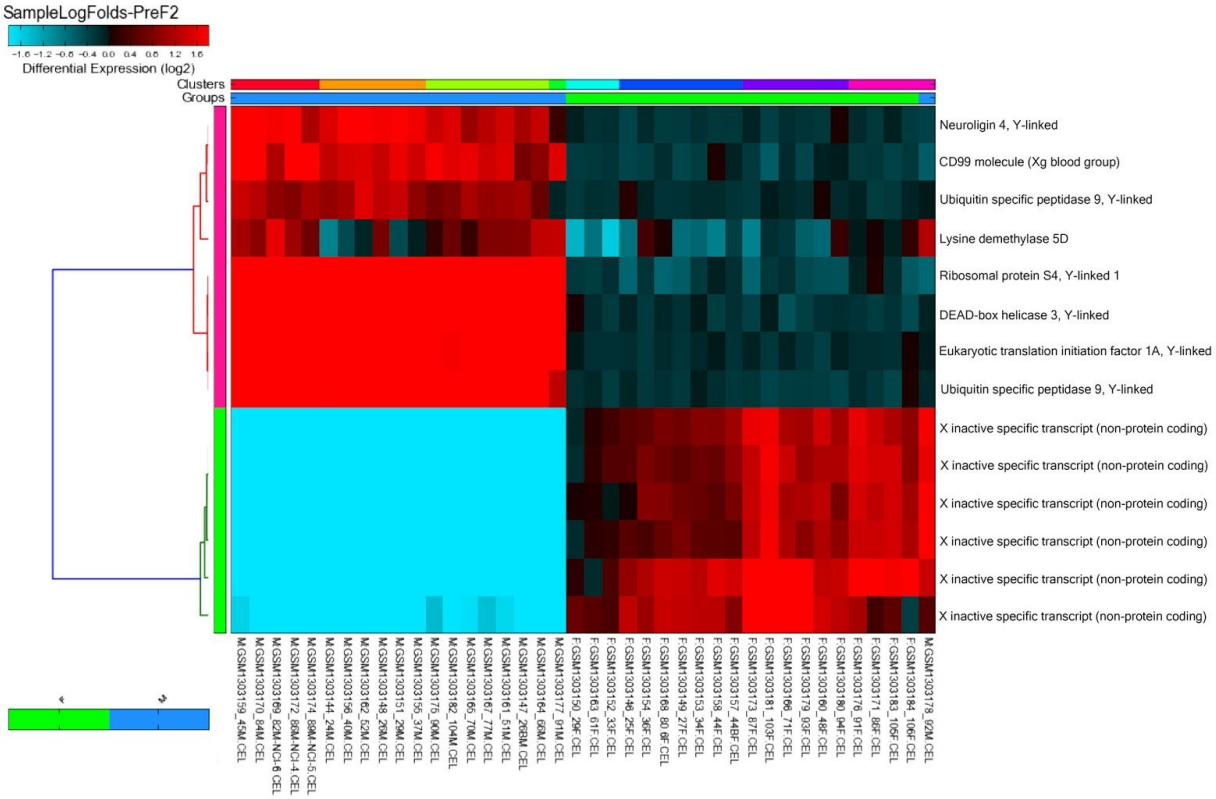

Figure 2. Data set heatmap with hierarchical clustering of rows and columns of all 14 significantly differentially expressed genes.

The cluster analysis heatmap, generated by AltAnalyze, contains all fourteen differentially regulated genes. The genes for the CD99 molecule and lysine demethylase break the expected pattern of X linked genes being down regulated in men and up regulated in women and y linked genes being up regulated in men and down regulated in women.

Principle Component Analysis:

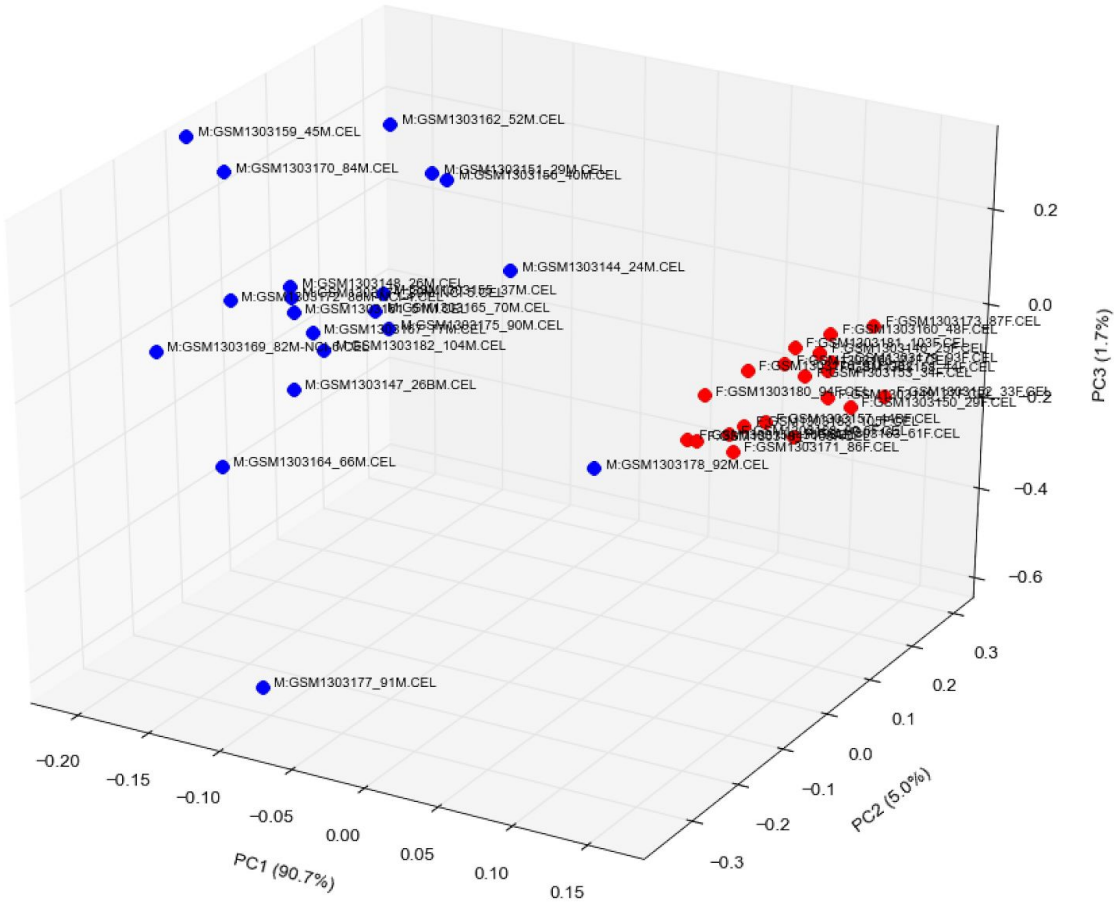

Figure 3. 3D PCA

All the variance between samples can be attributed to whether the patient was male or female. The male and female samples clustered together into two groups. With a PC1 of 90.7 percent, the pathway analysis is significant. There is a clear separation between male and female samples, with the exception of a couple outliers.

Pathway Analysis:

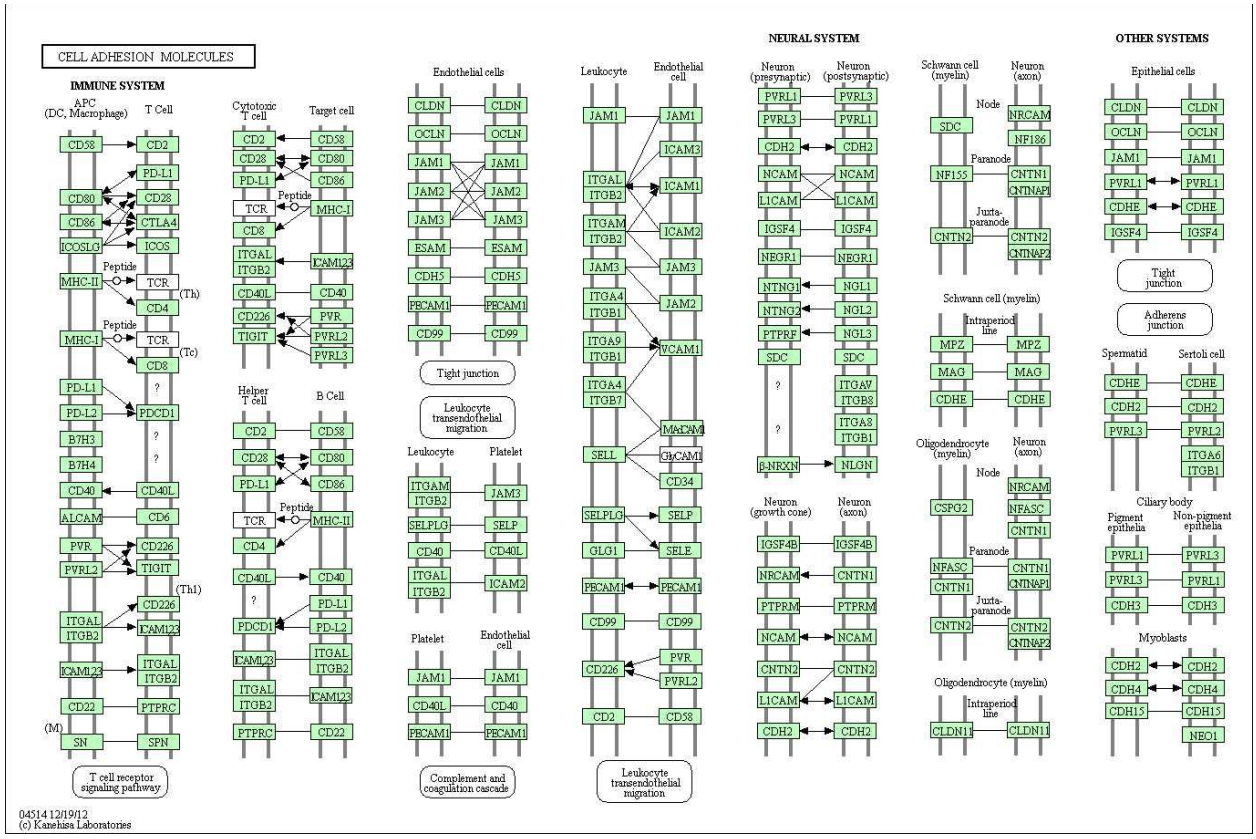

Figure 4. Genes involved in cell adhesion molecules.

DAVID pathway analysis shows that CD99 is involved in cell adhesion between endothelial cells and between leukocyte and endothelial cells. The adhesion between leukocyte and endothelial cells is important in the transendothelial migration of leukocyte cells, as the second pathway diagram shows.

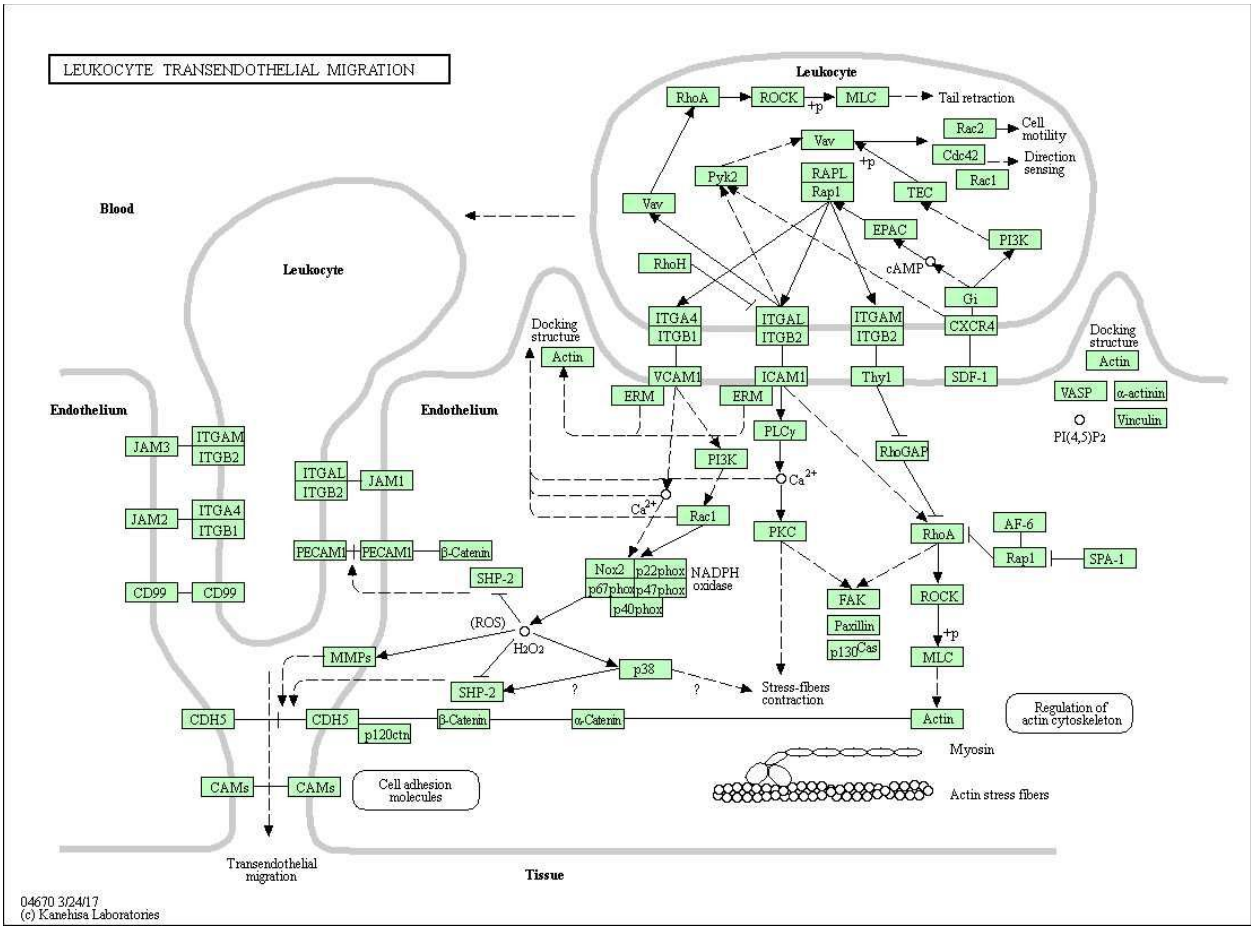

Figure 5. Genes involved in Leukocyte Transendothelial Migration

In endothelial cells, CD99 is expressed on the surface and plays an important role in Leukocyte migration. The forward migration of leukocyte cells between the junctions of endothelial cells (diapedesis) has been shown to be regulated by CD99 in conjunction with other molecules (PECAM-1). CD99 functions solely in diapedesis and plays no earlier role in endothelial to leukocyte adhesion during leukocyte migration (4). CD99 was shown by Watson et al. to have a role in LBRC recycling and leukocyte transendothelial migration. CD99 is found in at the endothelial border in an association with sAC, the scaffolding protein ezrin, and protein kinase A (PKA). Engagement of CD99 causes activation of Sac and release of cAMP. PKA is activated, followed by LBRC recycling and leukocyte transendothelial migration (1).

Figure 6. Global Pathway

Conclusion:

While fourteen genes were identified as being differentially regulated between men and women, twelve of the fourteen were either x or y linked. We chose to only consider CD99 because it is encoded by the *MIC2* gene located in the pseudoautosomal region on the X and Y chromosomes, it has also been correlated with many types of cancer including: Ewing Sarcoma, lymphoblastic lymphoma/leukemia, and malignant gliomas (1).

According to our results, CD99 is upregulated in men, which is concurrent with previous studies (5). It has been shown to be highly expressed in brains with malignant glioma and is involved with the migration of tumor cells (6). Statistics show that brain cancer is more common in men than in women. According to the National Cancer Institute, from 2011 to 2015, the number of new cases of brain and other nervous system cancers per 100,000 people was 7.5 for men and 5.4 for women (7).

Further research could shine light on the link between expression of CD99 in the brain and the development of cancerous brain tumors, as well as the mechanism of CD99 upregulation in men.

References:

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